SEMINAR REPORT

A seminar Report on

AI-Based Automatic UI/UX Design Generator Using NLP

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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DEPARTMENT OF COMPUTER ENGINEERING

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ABSTRACT

Designing a user interface (UI) and user experience (UX) is a time-consuming task that normally needs design skills, suitable tools and repeated revisions. Students, beginners and small teams often find it difficult to turn a written idea or requirement into a clear interface prototype. Manual and drag-and-drop design tools reduce some effort, but the designer still has to arrange every screen and component.

This seminar presents the AI-Based Automatic UI/UX Design Generator using Natural Language Processing (NLP). The system accepts a plain-English description of an application, interprets the user's intent, extracts the required screens, components and style preferences, and prepares a structured UI specification. A generation module then produces a wireframe or front-end code, which the user can preview and refine through follow-up prompts. The report reviews twelve research papers on text-to-UI retrieval and generation, conversational customization, prompt improvement and UI-to-code generation. The study shows that such systems can speed up early prototyping, while ambiguous prompts, limited design data and the need for human review remain important limitations.

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CHAPTER 1

INTRODUCTION

Mobile and web applications are used in education, banking, healthcare, shopping and many other areas, and the quality of their user interface (UI) and user experience (UX) strongly affects how easily people can use them. Designing an interface normally involves understanding requirements, sketching screens, choosing components and layouts, building prototypes and revising them after feedback. This process takes time and needs design experience.

Artificial Intelligence (AI) and Natural Language Processing (NLP) offer a new way to begin this process. Instead of drawing every screen, a user can describe the required application in ordinary language, for example, "a food delivery app with a login screen, a restaurant menu, a cart and a payment page". The AI-Based Automatic UI/UX Design Generator interprets such a description and converts it into an interface design.

The proposed system follows a pipeline approach. NLP techniques identify the intent of the prompt and extract entities such as the application type, screens, components and style preferences. If the prompt is incomplete, the system can ask clarifying questions. The extracted information is converted into a structured UI specification, which is used to select components, plan the layout and generate a wireframe or front-end code. The user can then preview the result and give follow-up instructions to refine it.

Recent research supports this approach. Existing studies show how text can be used to retrieve or generate UI mock-ups, how conversational instructions can modify interfaces, how prompts can be improved using user-preference knowledge, and how layout structure and visual feedback can improve generated code. However, these studies address different parts of the problem, and generated designs still need human review.

The main objective of the AI-Based Automatic UI/UX Design Generator is to reduce the time and skill needed to create an initial interface design while keeping the user in control of the final result. Its effectiveness can be judged by how well the output matches the prompt, its usability and accessibility, its visual consistency and how easily it can be edited.

CHAPTER 2

LITERATURE REVIEW

1. Creating User Interface Mock-ups from High-Level Text Descriptions with Deep-Learning Models (Huang et al., 2021)

This paper studies how deep-learning methods can create UI mock-ups from high-level text descriptions. It examines a text-only retriever, a multi-modal retriever and a UI generator that can find similar interfaces or generate a candidate mock-up to support early prototyping.

2. Data-Driven Prototyping via Natural-Language-Based GUI Retrieval - RaWi (Kolthoff et al., 2023)

RaWi helps users prepare GUI prototypes by searching a large repository of mobile-app screens using natural-language descriptions. It ranks matching screens with information-retrieval and BERT-based models and lets users edit the retrieved screens in a prototyping environment.

3. AI-Assisted UI Design: Enhancing Efficiency and Creativity through Generative Tools (Sun, 2026)

This paper explores how generative AI can convert natural-language prompts and low-fidelity sketches into editable interface layouts. In a user study, AI assistance reduced the reported workload and gave higher usability scores than the manual process, although ambiguous prompts sometimes needed substantial correction.

4. Towards Human-AI Synergy in UI Design: Enhancing Multi-Agent Based UI Generation with Intent Clarification and Alignment (Yuan et al., 2024)

This paper presents a human-centred, multi-agent workflow called PrototypeAgent for UI generation. It uses prompt augmentation and specialized agents to clarify what the user wants, generate UI components and support iterative refinement of the intermediate results.

5. Crafting User-Centric Prompts for UI Generations Based on Kansei Engineering and Knowledge Graph (Feng et al., 2025)

This paper improves prompts for UI-generation models by combining text mining, Kansei Engineering and knowledge graphs. Important terms in a user's description are linked to structured design knowledge, and a case study on automotive human-machine interfaces illustrates the approach.

6. NLDesign: A UI Design Tool for Natural Language Interfaces (Zhang et al., 2024)

NLDesign is a tool that lets users design and customize interfaces through natural-language instructions. Users can request changes to layouts, component properties and interaction logic, inspect the result and refine it with follow-up instructions.

7. Generative and Malleable User Interfaces with Generative and Evolving Task-Driven Data Model (Cao et al., 2025)

This paper proposes generative and malleable interfaces built around a task-driven data model. A language model converts the user's prompt into a task model, which is mapped to UI specifications and rendered, and users can then modify the interface using language or direct manipulation.

8. MorphGUI: Real-time GUIs Customization with Large Language Models (Calo et al., 2026)

MorphGUI allows people to customize an existing graphical user interface using ordinarylanguage requests. The framework changes existing components or creates new ones, and it was tested in a calendar-app use case with a user study of 18 participants.

9. DesignCoder: Hierarchy-aware and Self-correcting UI Code Generation with Large Language Models (2026)

DesignCoder converts structured UI mock-ups into front-end code while preserving the layout hierarchy. It identifies functional regions and component trees and uses a vision-guided selfcorrection loop that compares the rendered output with the design and repairs defects.

10. MLLM-Based UI2Code Automation Guided by UI Layout Information - LayoutCoder (Wu et al., 2025)

This paper generates UI code from real-world webpage images by first parsing the page into a layout tree that shows component relationships. The layout information guides code generation, and the authors also introduce the Snap2Code dataset of 350 websites for evaluation.

11. UI2Code^N: A Visual Language Model for Test-Time Scalable Interactive UI-to-Code Generation (Yang et al., 2025)

UI2Code^N is a visual language model for interactive UI-to-code generation. It uses staged training and a multi-turn workflow with visual feedback, and it supports generating code from a UI, editing an existing UI and polishing its appearance.

12. AI-Driven UX/UI Design: Empirical Research and Applications in FinTech (Xu et al., 2024)

This study examines the use of AI in UX/UI design for financial technology applications using surveys, interviews, usability tests and text analysis. It reports benefits such as personalization and engagement, but also highlights privacy, security and algorithmic-bias challenges.

Table No 2.1: Literature review:

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Retrieval and generation of UI mock-ups from text	Text-only retriever, multi-modal retriever, UI generator (deep learning)	Not specified	Turn high-level text descriptions into preliminary UI mockups	Depends on prompt clarity and available interface data; output may need manual correction
2	Natural-languagebased GUI retrieval for prototyping	Information retrieval, BERT-based learningto-rank, web prototyping editor	Rico repository of mobile-app screens	Find and reuse existing app screens from text queries to speed up prototyping	Retrieved screens may not match exactly; depends on repository coverage; supports early requirements stage only
3	Generative prototyping from prompts and sketches	Sketch recognition, natural-language prompts, generative tools, UI component constraints	UISketch (about 19,000 sketches, 21 categories, 1,218 participants)	Reduce prototyping time and designer workload while offering more layout alternatives	Ambiguous or specialized prompts can need substantial correction; risk of generic designs
4	Multi-agent UI generation with intent clarification	Prompt augmentation, specialized agents (PrototypeAgent)	Not specified	Clarify user intent and refine generated UI components step by step	Depends on the underlying language models and clarity of instructions; arXiv preprint
5	Prompt improvement using design knowledge	Text mining, Kansei Engineering, knowledge graph	User reviews and design knowledge; automotive HMI case study	Create better prompts so that generated UIs reflect user preferences	Depends on quality and coverage of review and knowledge data; needs testing on other app types
6	Conversational UI customization	Generative AI, natural-language instructions	Not specified	Modify layouts, component properties and interaction logic through language	Vague or conflicting instructions may be misread; result needs usability and accessibility review
7	Task-model-driven generative UI	LLM-generated task model, UI specifications, rendering	Not specified	Generate and evolve adaptable interfaces from natural-language requests	Depends on task-model quality and supported UI patterns; not equally good for all interface types
8	Natural-language customization of existing GUIs	Large language models; component modification and creation	Calendar-app use case; user study with 18 participants	Let users change the appearance or function of an existing GUI using ordinary language	Small study in one application; adapts existing GUIs rather than generating a full application
9	Structured mockup to code with self-correction	Multimodal LLM, UI grouping chain, vision-guided selfcorrection	300 high-fidelity mock-ups	Generate front-end code that preserves hierarchy and design details	Tailored to structured mock-ups; not tested for text-only input

10	Layout-guided image-to-code generation	Multimodal LLM, component relationships, UI layout tree	Snap2Code (350 real-world websites)	Preserve page layout when generating code from webpage images	Image-to-code rather than text-first; depends on layout parsing and evaluation datasets
11	Interactive UI-tocode generation with visual feedback	Visual language model, staged training, multi-turn refinement	UI-to-code and UIpolishing benchmarks	Generate, edit and polish UI code through repeated feedback	Focused on visual input; needs adaptation for textonly prompts; depends on benchmarks
12	Empirical study of AI in UX/UI design (FinTech)	Surveys, interviews, usability testing, text mining	Survey of 5,000 users; 30 expert interviews; 100 test participants	Study AI personalization, engagement and responsible design in financial apps	Privacy, bias and compliance challenges; findings limited to FinTech

CHAPTER 3

INFORMATION/CHAPTERS

1. Overview of AI UI/UX Design Generator:

An AI-Based Automatic UI/UX Design Generator is a software system that creates interface designs from natural-language descriptions. The user describes the required application, and the system converts the description into screens, components and layouts. Its main goal is to speed up early-stage UI/UX prototyping while keeping the user in control of the final design.

2. Role of Natural Language Processing (NLP):

Natural Language Processing (NLP) enables a computer to read and interpret human language. In the design generator, NLP is used to find the intent of the prompt (for example, creating a new application or changing an existing screen), to extract entities such as the application type, screen names, components, colours and layout preferences, and to detect missing or ambiguous details. The output of this stage is a structured set of requirements that the rest of the system can use.

3. Need for Intelligent Design Generation:

Manual UI design requires time, tools and experience, and many users cannot easily express their ideas visually. Retrieval-based tools only return existing screens, and simple template tools give limited control. An NLP-based generator can understand a description, ask for missing details and create a design that matches the request, while still allowing the user to refine the result. This reduces the effort needed to start a design and helps non-designers take part in the design process.

4. Working Principle:

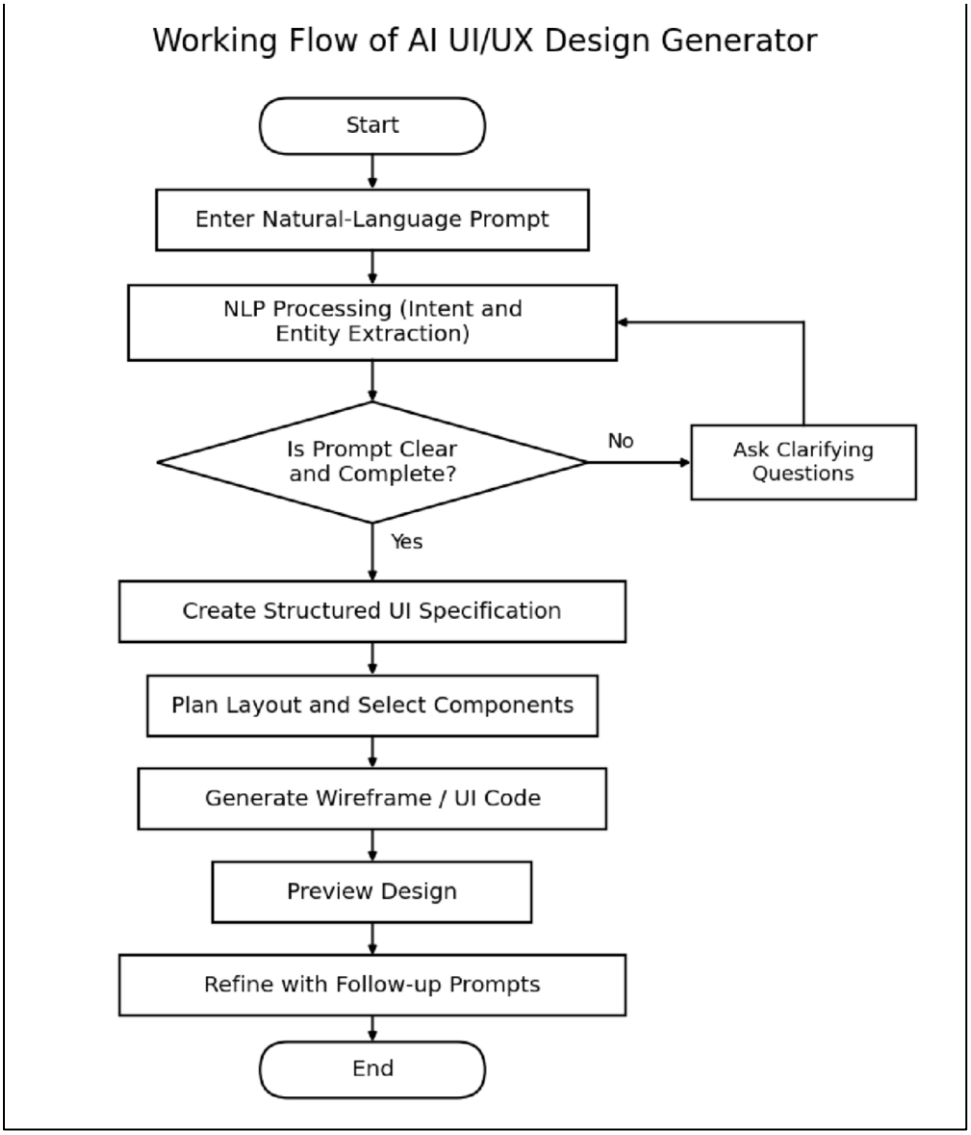


Fig 3.1: Working of AI UI/UX Design Generator

The working of the AI UI/UX Design Generator consists of the following steps:

1. **Start:** The process begins when the user wants to create an interface design.
2. **Enter Natural-Language Prompt:** The user describes the application in ordinary language, including its type, screens, features and preferred style.
3. **NLP Processing:** The system analyses the prompt to find the intent and extracts entities such as the application type, screen names, components, colours and layout preferences.

4. **Check Prompt Clarity:** The system checks whether the extracted information is enough to create a design.
5. **Ask Clarifying Questions:** If details are missing or unclear, the system asks the user questions, and the answers are processed again by the NLP module.
6. **Create Structured UI Specification:** The extracted information is converted into a structured specification that lists the screens, components and their relationships.
7. **Plan Layout and Select Components:** The system arranges the screens and chooses suitable components, such as buttons, forms, lists and navigation bars, using common design patterns.
8. **Generate Wireframe / UI Code:** A wireframe or front-end code is generated from the specification.
9. **Preview Design:** The generated design is shown to the user so that it can be checked against the original requirement.
10. **Refine with Follow-up Prompts:** The user asks for changes in ordinary language, for example, "make the buttons larger", and the system updates the design. The final design can then be exported.
11. **End:** The process is completed when the user accepts the design.

5. System Architecture and Main Components:

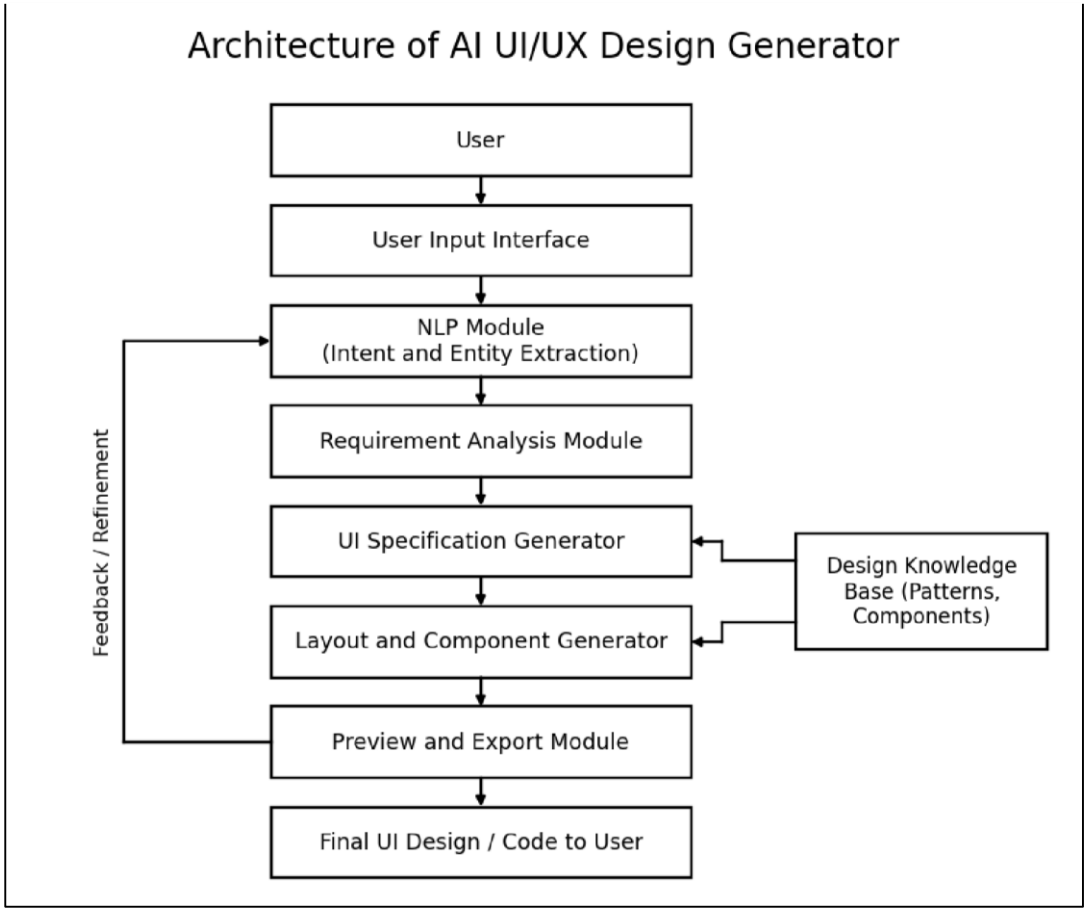


Fig 3.2: Architecture of AI UI/UX Design Generator

The main components include the User Input Interface, NLP Module, Requirement Analysis Module, UI Specification Generator, Layout and Component Generator, and Preview and Export Module. A Design Knowledge Base containing common UI patterns and components supports the generation stages. The NLP Module extracts intent and entities from the prompt, the Requirement Analysis Module checks for missing details, and the UI Specification

Generator prepares a structured description of the screens. The Layout and Component Generator then produces the wireframe or code, and the Preview and Export Module shows the result and passes the user's feedback back to the NLP Module for refinement.

6. Technologies and Techniques Used:

Python can be used to develop the system. NLP libraries such as spaCy can perform tokenization, intent detection and entity extraction, while a large language model accessed

through an API can help interpret complex prompts and generate UI specifications or code. A structured format such as JSON can store the intermediate UI specification, for example, a list of screens with their components and styles. HTML, CSS and JavaScript can be used to render the generated interface for preview, and a design tool such as Figma can be used as a reference or for optional export. The final technology selection depends on the project implementation and available resources.

7. Example of Prompt-to-UI Generation:

Consider the prompt: "Create a food delivery app with a login screen, restaurant list, menu page, cart and payment page. Use an orange colour theme and a simple modern style."

The NLP module identifies the application type (food delivery), the required screens (login, restaurant list, menu, cart and payment) and the style preferences (orange theme, simple and modern). The system then creates a specification and places suitable components, such as a login form, a scrollable restaurant list, item cards with an Add button, a cart summary and a payment form, to generate the connected screens. If the user later asks, "Make the Order button larger and move the cart icon to the bottom bar", the system updates only the requested parts of the design.

8. Preview, Refinement and Human Review:

The generated design is shown in a preview so that the user can check whether it matches the requirement. The user can give follow-up prompts to change layouts, components or styles until the design is acceptable. Research shows that generated interfaces can misunderstand ambiguous instructions, so the output should be treated as a starting point. A human designer or developer should still review the design for relevance to the prompt, usability, accessibility, visual consistency and correct behaviour.

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Faster Prototyping: Converts a written description into an initial design quickly, reducing the time needed to start a project.

2. Easy for Beginners: Allows students and non-designers to create interface drafts without advanced design skills.

3. Reduced Cost and Effort: Reduces manual work in sketching screens and arranging components.

4. Iterative Refinement: Supports follow-up prompts so the design can be improved step by step.

5. Consistent Design: Uses common patterns and components to keep layouts and styles consistent across screens.

4.2 Disadvantages:

1. Ambiguous Prompts: Vague or conflicting descriptions may produce designs that do not match what the user wanted.

2. Limited Creativity: Generated layouts may follow common patterns and lack originality.

3. Need for Human Review: Output must be checked for usability, accessibility and correct behaviour before use.

4. Dependence on Models and Data: The quality of results depends on the language model and the available design data and patterns.

5. Limited Support: Complex or highly specialized interfaces may not be handled equally well by the generator.

CHAPTER 5

CONCLUSION

The AI-Based Automatic UI/UX Design Generator provides a way to turn natural-language descriptions into interface designs. By using NLP to understand intent and extract requirements, it can create a structured specification and generate wireframes or front-end code, which reduces the time and effort needed to produce an initial design.

The reviewed papers cover different parts of this process: retrieval and generation of mock-ups from text, intent clarification, task-based interface generation, conversational customization, prompt improvement, and code generation using layout understanding and visual feedback. Together they suggest that an effective generator should interpret user intent, produce a suitable first design, allow iterative refinement and keep human review in the process. Challenges such as ambiguous prompts, model reliability and limited design data remain, and the reviewed studies do not show that production-ready results are guaranteed for every task.

In conclusion, an AI-based UI/UX design generator can be a useful assistant for early-stage design, helping users explore ideas faster, while designers and developers remain responsible for usability, accessibility and the final quality of the interface.

CHAPTER 6

PROJECT INITIATION

6.1 Description of Problem Statement:

Creating a user interface from an idea needs design knowledge, suitable tools and a lot of manual effort. Students, beginners, small businesses and developers without a design background often find it difficult to convert written requirements into screens and layouts. Even for experienced designers, producing several versions of an interface is time-consuming.

Existing approaches have limits. Retrieval-based tools only find existing screens that may not match the requirement, some generation tools depend on very clear prompts, and many UI-tocode methods need a mock-up or screenshot as input instead of a text description. Ambiguous prompts can also lead to designs that do not match what the user wanted.

Therefore, there is a need for a system that accepts a natural-language description, understands the user's intent, asks for missing details, generates a suitable UI design and allows refinement through follow-up prompts. The proposed AI-Based Automatic UI/UX Design Generator aims to address this need by combining NLP-based requirement extraction, structured UI specification and automatic layout generation, while keeping the user in control of the final design.

6.2 Scope and Objectives:

The scope of the project is to develop a system that generates user-interface designs from natural-language descriptions. The system will interpret the prompt using NLP, extract screens, components and style preferences, create a structured UI specification and generate a wireframe or front-end code for web or mobile screens. Users will be able to preview the design and refine it using follow-up prompts. The project focuses on early-stage prototyping and does not aim to build complete, production-ready applications or to replace professional designers.

Objectives of the Project:

1. To study AI and NLP techniques used to generate or customize user interfaces.

2. To interpret natural-language prompts and extract intent, screens, components and style preferences.

3. To convert the extracted requirements into a structured UI specification.
4. To generate wireframes or front-end code from the specification.

5. To allow iterative refinement of the design through preview and follow-up prompts.

6. To evaluate generated designs for relevance to the prompt, usability, accessibility and consistency.

6.3 Stakeholders:

The stakeholders of the AI-Based Automatic UI/UX Design Generator project are the individuals and groups who use, develop, maintain, or benefit from the system.

1. **End Users:** Students, beginners and small business owners who want to create interface designs without advanced design skills.

2. **UI/UX Designers:** Professionals who can use the generator to create quick first drafts and explore alternative layouts.

3. **Developers:** The team responsible for designing, developing, testing and maintaining the generator, and developers who use the generated layouts or code.

4. **Clients and Organizations:** Businesses and institutions that need quick prototypes to discuss and confirm requirements.

5. **AI Service Providers:** Organizations that provide the language models or APIs used for prompt understanding and generation.

6. **Project Guide and Evaluators:** Faculty members who guide the development process and evaluate the project's design, implementation and results.

6.4 Platform / Equipment / Resources:

1. Platform:

The AI-Based Automatic UI/UX Design Generator can be developed on the Windows or Linux operating system using Python for the NLP pipeline and HTML, CSS and JavaScript for rendering the generated interface preview. A code editor such as Visual Studio Code can be used for development and testing. The system may connect to a language model through an API to interpret prompts and generate UI specifications or code.

2. Equipment:

- **Computer or Laptop:** For developing, running and testing the application.
- **Processor:** A standard modern processor suitable for Python development.
- **RAM:** Minimum 8 GB recommended; additional memory may be required if language models are run locally.
- **Internet Connection:** Required for accessing language-model APIs and downloading libraries.
- **Storage:** Sufficient space for source code, libraries, datasets and generated design files.

3. Resources:

Table No 6.1: Resources:

Resources	Name	Purpose
Operating System	Windows 11	Host and run development tools, editors and the local testing environment.
Programming Language	Python 3.11	Implement the NLP pipeline, backend logic, scripts and tests.
Code Editor	Visual Studio Code	Code editing, debugging and extension support for development.
NLP Library	spaCy	Tokenization, intent detection and entity extraction from user prompts.
Language Model / AI Service	OpenAI API	Prompt understanding and generation of UI specifications or code.
Front-End Technologies	HTML, CSS, JavaScript	Render the generated UI as a live preview in the browser.
Design Tool	Figma	Reference for layouts and optional export of generated designs.
Version Control	Git and GitHub	Track code changes and support team collaboration.

CHAPTER 7

Seminar Presentation

S.S.V.P.S's B. S. Deore Polytechnic, Dhule Department of Computer Engineering *Academic Year : 2025-2026*	
SEMINAR AND PROJECT INITIATION COURSE	
Sem : CO 5 K	**SPI – 315003**	**Date : 11 Oct' 2025**
Topic : AI-Based UX/UI Design Generator Using NLP	
Roll No.	**Name of Student**
56	Jagruti Pankaj Patil
53	Hemangi Dilip Patil
45	Ishwari Vikas Nagare
42	Sarthak Dnyaneshwar Marathe

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Abstract

AI-Based UX/UI Design Generator Using NLP is a proposed system designed to simplify the process of user interface and user experience design. Traditional UI/UX design requires considerable time, effort, and design knowledge. The proposed system uses Natural Language Processing (NLP) to understand user requirements written in simple language and identify important interface components such as pages, buttons, forms, layouts, and content.

It aims to assist in generating an initial interface design, reduce manual effort, and improve the efficiency of the prototyping process. The system can help developers and users create initial UI designs without requiring advanced design skills. The generated designs can be reviewed and modified according to user requirements.

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2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	NLP-based requirement extraction	NLP, GUI Retrieval, BERT-based models	GUI design repository	Retrieve suitable GUI designs based on user requirements.	Retrieved designs may require manual modifications.
2	AI-assisted UI/UX design	Artificial Intelligence, Machine Learning	Depends on the research study	Improve design efficiency and automate repetitive tasks.	Output quality depends on the AI model.
3	Generative UI design	Generative AI, Large Language Models	Depends on the model and evaluation method	Generate interface layouts and components from user prompts.	Generated layouts may not fully match user requirements.

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Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	NLP-based requirement extraction	NLP, Text Processing, Tokenization	Natural-language UI requirement	Identify pages, buttons, forms and other UI components.	Ambiguous text may produce incorrect results.
5	Text-to-interface generation	NLP, Generative AI, HTML, CSS	Text prompts and generated interfaces	Convert text descriptions into initial UI designs.	Complex prompts may generate incomplete interfaces.
6	Data-driven GUI prototyping	GUI Retrieval, NLP, Information Retrieval	Existing GUI examples	Reuse relevant interface examples to reduce prototyping effort.	Retrieved examples may need customization.

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Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	AI-based layout generation	Generative AI, Machine Learning	UI layouts and design examples	Generate suitable layouts based on user requirements.	Layout consistency may vary.
8	Automated UI component generation	AI, NLP, HTML, CSS	UI component examples	Generate buttons, forms and other interface elements.	Advanced styling may require manual work.
9	Human-AI collaborative design	AI design tools, Generative AI	Designer feedback and interface examples	Combine AI-generated designs with human creativity.	Human review remains necessary.

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Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	AI-assisted rapid prototyping	Generative AI, UI design tools	Prototype examples	Reduce the time required to create initial prototypes.	Results may need repeated refinement.
11	NLP-based UI customization	NLP, Machine Learning	User requirements and interface examples	Adapt interface components according to user needs.	Complex requirements may not be handled accurately.
12	Integrated AI-based UI/UX generation	NLP, Generative AI, Web Technologies	Depends on the implemented system	Combine requirement understanding and interface generation.	Accuracy and customization depend on system capabilities.

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3. Information to seminar topic

- **Technology:** Artificial Intelligence (AI), Natural Language Processing (NLP), Generative AI and Web Technologies.
- **Main Purpose:** To simplify UI/UX design by converting natural-language requirements into an initial interface design.
- **Key Concept:** Understanding user requirements and identifying suitable UI components such as buttons, forms, pages and layouts.
- **Approach:** The proposed system processes the user's text, extracts important requirements and assists in generating an initial interface design.
- **Main Benefit:** Reduces manual effort, saves time and supports rapid prototyping.
- **Applications:** Website prototyping, mobile application design, educational projects, business applications and software development.

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4. Advantages / Disadvantages

Advantages:

- Reduces the time required for initial UI/UX design.
- Allows users to describe requirements in natural language.
- Automates parts of the interface design process.
- Supports rapid prototyping and design experimentation.
- Reduces repetitive manual work.
- Can store relevant project data using MongoDB, if implemented.

Disadvantages:

- Output quality depends on the clarity of user input.
- Complex requirements may not be interpreted correctly.
- Generated designs may require manual modifications.
- NLP may not extract every requirement accurately.
- Advanced customization may be limited.
- Human review is necessary before finalizing the design.

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5. Conclusion

AI-Based UX/UI Design Generator Using NLP simplifies the process of interface design by converting natural-language requirements into initial UI designs. It reduces manual effort, saves time, and supports rapid prototyping. Overall, it makes UI/UX design easier and more accessible, although human review and modifications may still be required.



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6. Project Initiation

The proposed AI-Based UX/UI Design Generator Using NLP addresses the time and effort required to create initial user interfaces. Traditional design methods require users to manually create layouts, buttons, forms and navigation elements.

The proposed system accepts requirements written in natural language and processes them using NLP techniques. It identifies important interface components and assists in preparing an initial design that can be reviewed and modified.

The project aims to simplify the prototyping process, improve productivity and make initial interface design easier for developers and users.

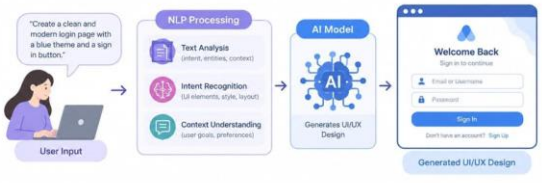
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6(a) Problem Statement

Traditional UI/UX design requires considerable manual effort, time and design knowledge. Users must manually plan layouts, forms, buttons and navigation elements to create an interface.

This process can be difficult for beginners and may require repeated modifications. The proposed system aims to address this problem by processing natural-language requirements and assisting in the generation of an initial interface design.

It focuses on reducing repetitive work, simplifying prototyping and improving the efficiency of the initial design process.



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6(b) Scope and Objective

Objectives:

Accept user requirements written in natural language.

Process text using Natural Language Processing (NLP).

Identify important UI components and design requirements.

Assist in generating an initial user interface.

Provide a simple and user-friendly design process.

Store relevant information using MongoDB, if included in the implementation.

Scope:

Initial UI/UX design generation from text requirements.

Identification of pages, buttons, forms and other components.

Creation of interface prototypes for review and modification.

Support for website and application prototyping.

Future integration with advanced Generative AI models.

Further improvements in customization and requirement understanding.

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6(c) Stake holders

- **End Users:**
  Provide requirements in natural language and review the generated interface.

- **UI/UX Designers:**
  Use AI-generated initial designs as a starting point for prototyping.

- **Developers:**
  Review the generated interface and use it during application development.

- **Project Team:**
  Develop, test and maintain the proposed system.

- **Project Guide:**
  Provides guidance, reviews project progress and evaluates the project.

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6(d) Platform / Equipment / Resources identification

1. Development Platform

Windows or Linux operating system.
Python development environment.
Web browser for viewing and testing interfaces.
Development tools for implementing NLP-based processing.

2. System Equipment

Computer or laptop for development and testing.
Minimum 4 GB RAM, with higher memory recommended for AI tools.
Internet connection for accessing documentation and AI services.
Keyboard and mouse for development work.

3. Supporting Resources

Python programming language.
NLP libraries and tools.
MongoDB for data storage, if implemented.
HTML and CSS for interface development.
Visual Studio Code or a similar code editor.
Research papers and technical documentation related to AI and UI/UX design.

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- Artificial Intelligence: A Modern Approach — Stuart Russell and Peter Norvig.
- Software Engineering — Ian Sommerville.
- Reference materials on user interface and user experience design.

Websites

- Python Official Website – https://www.python.org
- Python Documentation – https://docs.python.org/3
- MongoDB Official Website – https://www.mongodb.com
- Natural Language Toolkit (NLTK) – https://www.nltk.org

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Name of student
1	24610960221	Sarthak Dynaneshwar Marathe
2	24610960224	Ishwari Vikas Nagare
3	24610960232	Hemangi Dilip Patil
4	24610960235	Jagruti Pankaj Patil

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388529	388532	388540	388543
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : ----------

(Course Expert/s)